Freeing `rustc`'s Understanding: Third-Party Rust Tooling

alona.page/talks/freeing-rustcs-understanding.pdf

Alona Enraght-Moony

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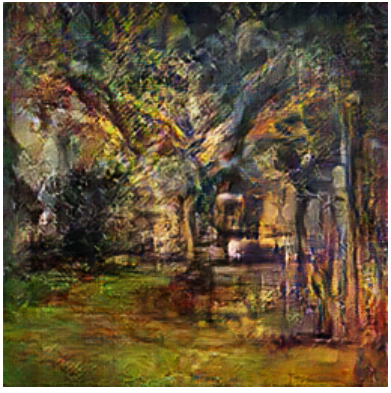
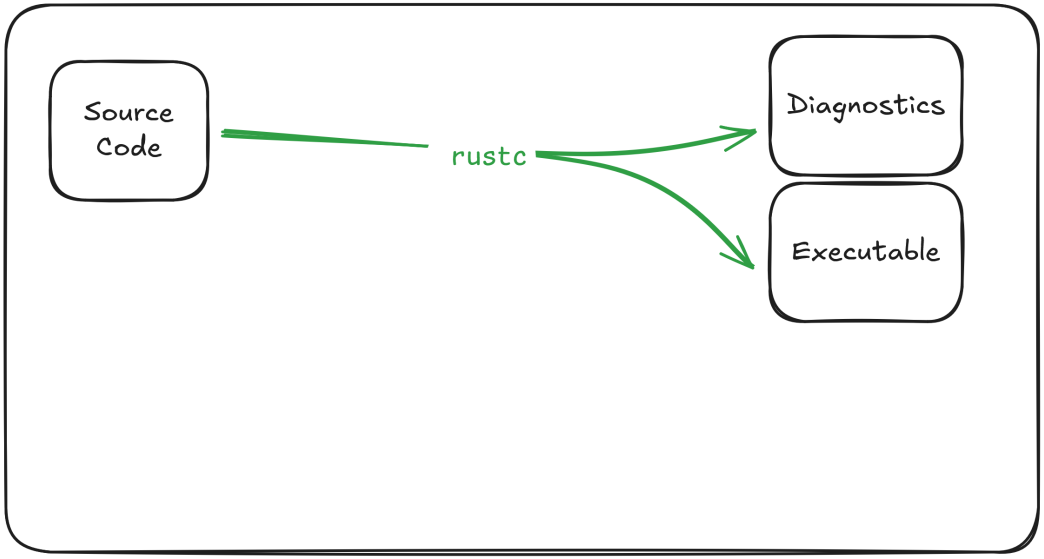
whoami(1)

- Alona Enraght-Moony (she/her)
- Rustdoc team since 2022
- Focused there on maintaining rustdoc-json
- This talk comes from soul-searching about the place that rustdoc-json fits into the ecosystem

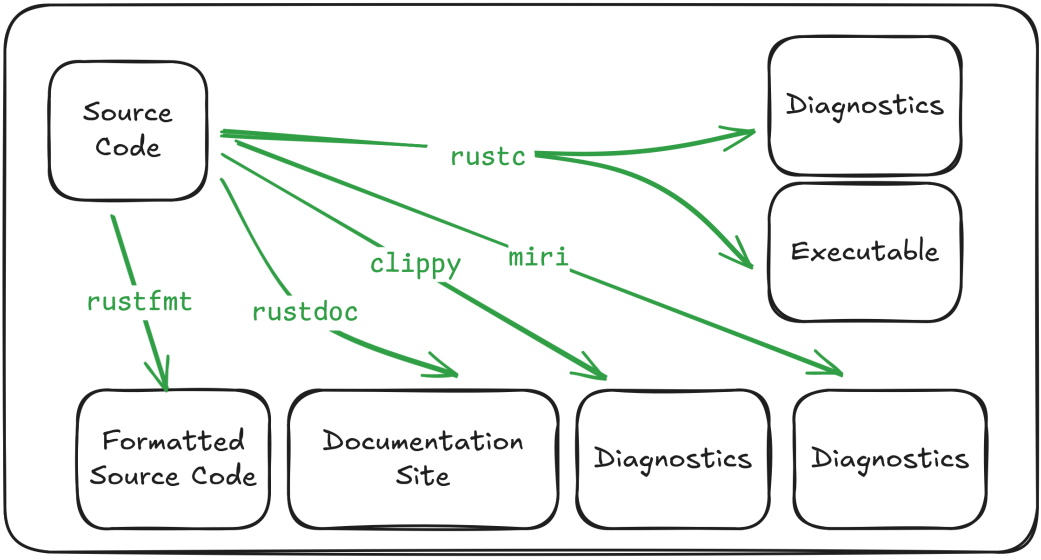


aDotInTheVoid

rustc, the 10'000 feet view



`rustc` isn't enough: First party tools



The rust project isn't enough: Third party tools

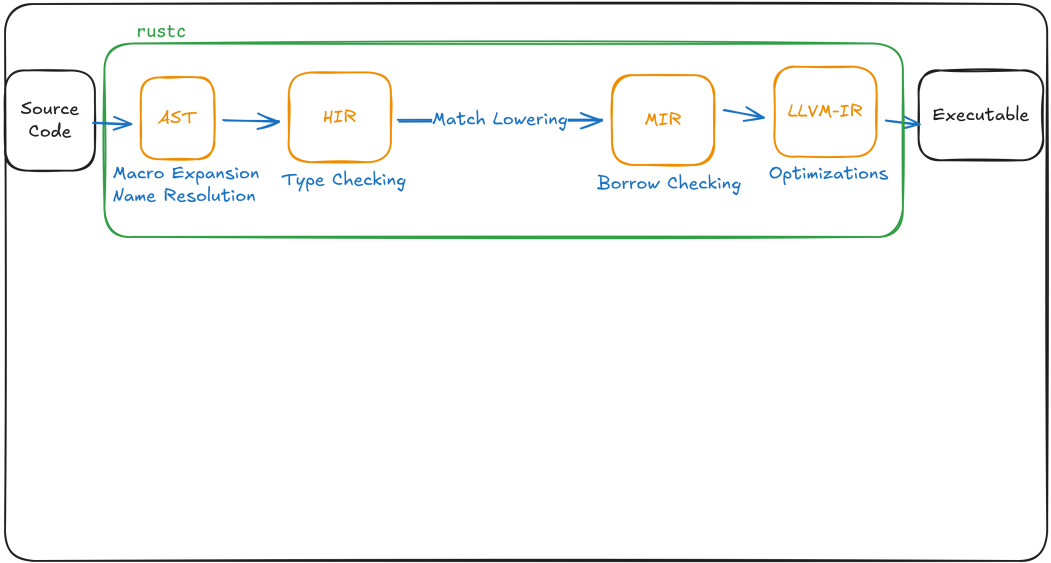
- Linters
  - cargo-check-external-types
  - cargo-semver-checks
  - klint
  - dylint
  - marker
- Verifiers
  - kani
  - aeneas
  - Creusot
- Refractoring tools
  - CoccinelleForRust
- Inspection
  - cargo-public-api
  - cargo-modules

Your tools here!

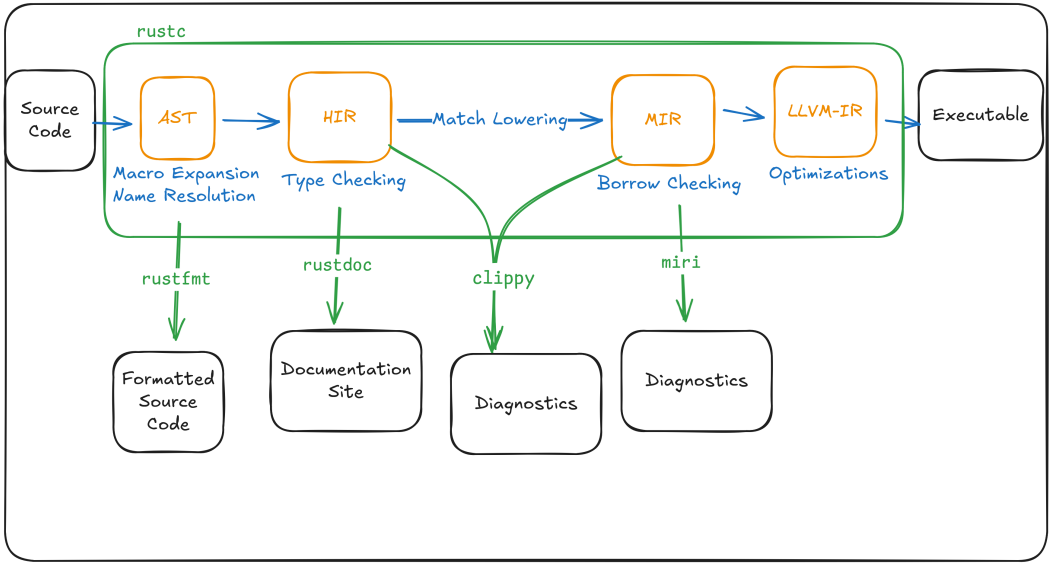
Agenda

- How tools work
- Why third-party tools are more challenging than first-party tools.
- Many different approaches to make it easier.

rustc, the 1'000 feet view



rustc, and the first party tools



How tools call into rustc

```
extern crate rustc_driver;
extern crate rustc_errors;

// set up target, flags, crate loader into `config`.

rustc_interface::run_compiler(config, |compiler| {
    let ast = rustc_interface::passes::parse(&compiler.sess);
    rustc_interface::create_and_enter_global_ctxt(compiler, ast, |tcx| {
        // real work here.
    })
});
```

Can third-party be like first party?

```
#![feature(rustc_private)]
extern crate rustc_driver;
extern crate rustc_errors;

// set up target, flags, crate loader into `config`.

rustc_interface::run_compiler(config, |compiler| {
    let ast = rustc_interface::passes::parse(&compiler.sess);
    rustc_interface::create_and_enter_global_ctxt(compiler, ast, |tcx| {
        // real work here.
    })
});
```

Can third-party be like first party?

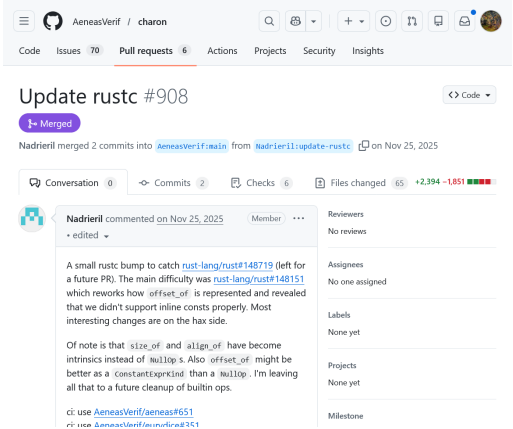
```
[toolchain]
channel = "nightly-2025-02-13"
components = ["llvm-tools-preview", "rustc-dev"]
```

Can third-party be like first party?

```
$ ldd ./target/debug/demo_tool
        linux-vdso.so.1 (0x00007ffff7fd6000)
        librustc_driver-2fb444505ec98d04.so => /home/alona/.rustup/toolchains/nightly-2026-01-18-x86_64-unknown-li
nux-gnu/lib/librustc_driver-2fb444505ec98d04.so (0x00007d2cf0a00000)
        libgcc_s.so.1 => /lib/x86_64-linux-gnu/libgcc_s.so.1 (0x00007d2cf86cd000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007d2cf0600000)
        libdl.so.2 => /lib/x86_64-linux-gnu/libdl.so.2 (0x00007d2cf86c8000)
        libLLVM.so.21.1-rust-1.94.0-nightly => /home/alona/.rustup/toolchains/nightly-2026-01-18-x86_64-unknown-li
nux-gnu/lib/../lib/libLLVM.so.21.1-rust-1.94.0-nightly (0x00007d2ce6800000)
        librt.so.1 => /lib/x86_64-linux-gnu/librt.so.1 (0x00007d2cf86c1000)
        libpthread.so.0 => /lib/x86_64-linux-gnu/libpthread.so.0 (0x00007d2cf86bc000)
        /lib64/ld-linux-x86-64.so.2 (0x00007d2cf8723000)
        libm.so.6 => /lib/x86_64-linux-gnu/libm.so.6 (0x00007d2cf0909000)
        libz.so.1 => /lib/x86_64-linux-gnu/libz.so.1 (0x00007d2cf869e000)
$
```

Problems for third-party tools

`rustc_private` APIs breaks frequently



Projects have died due to this:

- gazebo_lint
- semverver

`rustc_private` crates are unlike any other dependency

- `rustc` can't be build from crates.io with `cargo`, but needs it's own build system
- The version of `rustc` that builds your tool is also the version of `rustc` your tool uses
- You need to be on nightly to use them
- That same version of nightly needs to be installed at runtime

How the first-party tools avoid this

`rustc_private` APIs breaks frequently

Don't allow `rustc` changes to break first party tools.
Changes to `rustc` must make sure that `rustdoc`, `clippy`, `rustfmt`, `miri` tests still pass.

`rustc_private` crates are unlike any other dependency

- First party tools are built with `x.py`, not `cargo`
- First party tools are distributed by `rustup`, not `cargo`

Both of these are possible because first party tools are *developed, tested and distributed* alongside `rustc`.

Summary: `#![feature(rustc_private)]`

Lets you depend access the compiler internals.

- Pros:
  - Access to all the rich state rustc has
- Cons:
  - Large API, with frequent breaking changes
  - Only usable on nightly
  - Need to dynamically link to specific version of `librustc_driver-<hash>.so`

A level of indirection: `rustc_public`

A wrapper crate implemented using `#![feature(rustc_private)]`

Currently distributes with rustc as an `extern crate`

- Pros:
  - More limited API, designed for tools
  - Less breaking changes
  - One day, will be on `crates.io`, support multiple rustc versions.
- Cons:
  - Might not have the API you need, and need to fall back to `rustc_private` crates.
  - Still need to link to version specific `librustc_driver.so`
  - `crates.io` and multi-version are still hypotheticals (as of today)

rust-analyzer as a library

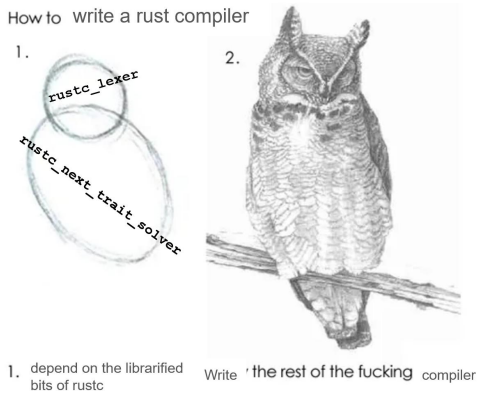
De nuevo implementation of a rust compiler, for IDE support.

- Pros:
  - Freestanding
  - Usable on stable
  - Use like any other library on crates.io
- Cons:
  - API changes frequently
  - Doesn't have a "complete" compiler
    - No control-flow-graph
    - No borrow-checking
  - API designed for interactive IDE tooling
    - Optimized for latency, not throughput

Behind (bits of) rust-analyzer: libraryification of rustc

Some bits of the compiler (e.g. `rustc_lexer` and `rustc_next_trait_solver`) are on `crates.io`, initially to be used by rust-analyzer.

- Pros:
  - Freestanding
  - Usable on stable
  - Use like any other library on crates.io
- Cons:
  - API's change sometimes
  - Only implements the "leaf"s of analysis, not the core.
    - Bring your own data-structures: `rustc_type_ir`
    - Draw the rest of the owl



The one I worked on: `rustdoc --output-format=json`

Rustdoc feature to emit an IR to a JSON file.

- Pros:
  - Talks to rustc over a json file, not memory
  - Your tool can be in stable, but use nightly rustdoc
  - Small API, infrequent breaking changes
- Cons:
  - Tied to `rustdoc`, not `rustc`: presentational representation, not semantic
  - Only describes API, nothing about the contents of functions
  - Can't request the information you need, get all the information whether you care or not

JSON Emiter but with function bodies: Charon

A third-party tool implemented with `#![feature(rustc_private)]` that emits an IR to a JSON file

- Pros:
  - Talks to rustc over a json file, not memory
  - Your tool can be in stable, but use nightly rustdoc
  - IR optimized for semantics and verification
- Cons:
  - Focused on semantics, not syntax.
  - Charon itself requires specific nightly.
  - Can't request the information you need, get all the information whether you care or not

Summary

- Compilers to more that just take inputs and produce outputs, their intermediates are essensial for building tools
- First party tools have convenient access to this intermediate state
- But third party tools do not
- Many approaches to getting at it

That's all folks!

Slides online: <alona.page/talks/freeing-rustcs-understanding.pdf>

Thanks

Jynn Nelson, Julia Ryan, David Barsky

Contact

- email: me@alona.page
- fedi: @adotinthevoid@hachyderm.io
- bluesky: alona.page